

Supply Chain Health Agent

Agentic AI-powered supply chain diagnostics with live market intelligence
Powered by Anthropic Claude SDK | Open Source — MIT License

v2.0 — May 2026

github.com/dwnjuguna/supply-chain-health-agent

8	8	3	2	Live	Auto
Health Domains	Verticals	Personas	Modes	Web Search	Action Pack

What's in this document

1. Product overview — what it does, who it's for, and what's new in v2.0
2. Three user personas — Analyst/Manager, C-Suite/Board, Enterprise (Phase 2)
3. Agentic capabilities — web search, auto action pack, market intelligence
4. The 8 health domains with 2025 world-class KPI benchmarks
5. 8 industry vertical presets with sector-specific benchmarks
6. Privacy, legal & compliance — GDPR, CCPA, illustrative use disclaimer
7. Project structure & source code summary
8. Quick-start guide — deploy in under 5 minutes
9. Phase 2 roadmap — cross-session memory, alerts, integrations
10. Links & contact

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1. PRODUCT OVERVIEW

The Supply Chain Health Agent is an agentic AI-powered diagnostic platform that evaluates an organisation's end-to-end supply chain across 8 critical domains. It benchmarks performance against 2025 SCOR and Gartner world-class standards, autonomously searches for live market intelligence, and delivers scored, prioritised recommendations — in under 60 seconds.

What's new in v2.0

■ Agentic web search	Claude autonomously searches for live supply chain benchmarks, commodity prices, regulatory updates, and geopolitical risks during every assessment — grounding analysis in real-time market data rather than static training knowledge.
■ Auto action pack	After every assessment, Claude autonomously generates three board-ready outputs without being prompted: a 5-bullet executive summary, a prioritised 90-day action plan with named owners and success metrics, and a top-5 risk watch list.
■ Live market intelligence	A dedicated tab surfaces real-time supply chain signals for the chosen vertical — freight conditions, commodity moves, regulatory changes, and geopolitical risks — updated every time an assessment runs.
■ Dual-track persona system	Users select their profile at launch. The Analyst/Manager track delivers deep KPI diagnostics. The C-Suite/Board track delivers strategic scenario comparison, a phased maturity roadmap, and board-ready language throughout.
■ Cost-benefit analysis	Optional financial inputs (revenue, inventory value, OTIF rate, gross margin) enable Claude to size the cost of each performance gap and estimate the ROI of closing it. Full GDPR/CCPA-compliant privacy controls — data never stored.
■ 2025 KPI benchmarks	Every domain now carries explicit world-class thresholds sourced from Gartner Top 25 2025, SCOR, and current market data — giving Claude precise numbers to score against rather than general guidance.

2. THREE USER PERSONAS

Users select their profile at the landing screen. The tool forks cleanly from entry — different inputs, different AI instructions, different output format and depth. A third persona (Enterprise Configuration) is designed and visible in the UI, shipping in Phase 2.

Persona	Track	Built for	Key outputs
■ Analyst / Manager	Practitioner	Supply chain managers, analysts, and operations teams who speak the KPI language	Domain scores, benchmark gaps, CBA, follow-up Q&A;
■ C-Suite / Board	Executive	CEOs, CFOs, COOs, and board members making investment and transformation decisions	Scenario comparison, maturity roadmap, board-ready summary, action pack
■■ Enterprise Config	Phase 2	Engineering and IT teams embedding the agent within internal systems	Admin panel, access controls, org presets, compliance guardrails, branding

Executive track — unique outputs

When a C-Suite or Board user runs an assessment, Claude produces two additional sections beyond the standard diagnostic:

Strategic Scenario Comparison

Three named strategic paths (e.g. 'Optimise Current Operations' vs 'Selective Transformation' vs 'Full Digital Transformation') — each with domains addressed, investment range, expected KPI movement, key trade-offs, and best-suited mandate. Closes with a recommended scenario and rationale.

Supply Chain Maturity Roadmap

A four-phase improvement plan: Phase 0 (Now), Phase 1 — Stabilise (0-6 months), Phase 2 — Build (6-18 months), Phase 3 — Lead (18-36 months). Each phase includes named initiatives, investment range, KPI milestone targets, and interdependencies. Closes with a total programme summary and cost-of-inaction statement.

Customisation options (executive track)

Parameter	Options
Strategic orientation	Balanced / Growth-first / Cost-out and efficiency-first
Timeline preference	Quick wins (0-12 months) / Medium-term / Long-term / All horizons
Benchmark peer	Gartner Top 25 / Industry average / Top quartile / Direct competitors
Report style	Board presentation / Executive briefing / Full strategic report

3. AGENTIC CAPABILITIES — PHASE 1

v2.0 moves beyond a conversational tool into a genuinely agentic system. Claude takes autonomous multi-step actions during every assessment — not just responding to prompts, but proactively searching, synthesising, and generating outputs without being asked.

Web Search Tool

Implemented via Anthropic's server-side `web_search_20250305` tool. Claude autonomously decides when and what to search during each assessment — fetching live SCOR benchmarks, freight indices, regulatory updates, and geopolitical risk signals. Up to 5 searches per assessment call, with graceful fallback to static knowledge if web search is unavailable.

Auto Action Pack

Triggered automatically after every assessment — no user prompt required. Claude generates three board-ready outputs: (1) a 5-bullet executive summary ready to paste into a board deck, (2) a prioritised 90-day action plan with suggested owners, success metrics, and week-by-week timing, and (3) a top-5 risk watch list with severity, likelihood, and immediate mitigation actions.

Live Market Intelligence

A separate Claude call searches for the 5 most important supply chain developments affecting the chosen vertical right now — covering freight and shipping conditions, commodity prices, regulatory changes, and geopolitical risks. Displayed in a dedicated Market Intelligence tab alongside the assessment results.

Agentic Follow-up Q&A;

The Q&A; loop now also has web search access. When a follow-up question requires current data ('What are freight rates doing right now?'), Claude can search autonomously before responding — rather than relying on potentially stale training data.

4. AGENTIC CAPABILITIES — PHASE 2 ROADMAP

Capability	What it means	Status
Cross-session memory	Claude remembers your organisation across sessions and tracks KPI trends over time	Phase 2
Proactive risk alerts	Claude monitors supply chain news and flags risks relevant to your vertical before you ask	Phase 2
Workflow integrations	Push summaries to Slack, create Jira tickets, draft emails — directly from the assessment	Phase 2
Enterprise configuration	Admin panel for IT teams to embed with custom branding, access controls, and org presets	Phase 2

5. THE 8 HEALTH DOMAINS — 2025 BENCHMARKS

Each domain is scored 0-100 against 2025 world-class thresholds sourced from Gartner Top 25, SCOR, and current market data. Claude uses these exact figures when scoring — not general guidance.

Domain	SCOR	Key KPIs	World-class target (2025)
Demand Planning & Forecasting	Plan	Forecast accuracy, bias, MAPE, S&OP; maturity	FA >85-90%, MAPE <10%, Integrated Business Planning
Procurement & Sourcing	Source	Supplier OTD, spend under management, PPM defects	OTD >95%, 80%+ spend managed, <500 PPM
Manufacturing & Production	Make	OEE, first-pass yield, schedule adherence	OEE >85%, FPY >98%, schedule adherence >95%
Inventory Management	Plan/Deliver	Inventory turns, stockout rate, E&O; %, C2C cycle	8-12x turns, stockout <1%, E&O; <2%
Logistics & Transportation	Deliver	OTIF, freight cost % revenue, perfect order rate	OTIF 95-98% (Walmart: 98%), freight <5% (B2C)
Warehousing & Fulfillment	Deliver	Order accuracy, utilization, lines/hour	Accuracy >99.9%, utilization 80-85%, AMR-enabled
Risk & Resilience	Enable	BCP coverage, TTR, supplier risk tiers, resilience index	BCP >95% nodes, TTR <2-4 weeks, <15% single-source
Sustainability & ESG	Enable	Scope 3 tracking, ESG audit coverage, CSDDD readiness	Scope 3 >80% tracked, >80% Tier-1 ESG audited

Score bands

Score	Rating	What it means
80-100	Excellent	World-class. At or above Gartner Top 25 benchmarks. Sustain and innovate.
60-79	Good	Above industry average. Clear improvement path — targeted action needed.
40-59	Fair	At or below industry average. Structured improvement programme required.
0-39	At Risk	Significantly below benchmarks. Immediate intervention required.

6. INDUSTRY VERTICAL PRESETS

Eight vertical presets adjust scoring benchmarks, risk weighting, domain emphasis, and reference organisations to reflect each sector's operational realities. General is now the 8th vertical (added in v2.0).

Vertical	Focus areas	Key benchmarks	Reference orgs
■ Semiconductor	Fab cycle times, die yield, export controls (CHIPS Act, BIS EAR), rare earth risk	Die yield >92%, fab utilisation 85-95%, critical material inventory 180+ days	TSMC, Intel, NVIDIA, ASML
■ Automotive	JIT/JIS delivery, zero line stoppages, EV transition readiness, PPM quality	PPM defects <50, line stoppage incidents = 0, IATF 16949 compliance	Toyota (TPS), BMW, Tesla
■ Pharmaceutical	Cold chain integrity, GDP compliance, FDA 21 CFR, DSCSA serialization	Cold chain compliance 100%, batch traceability 100%, recall readiness <2hrs	Pfizer, Roche, Johnson & Johnson
■ Retail & E-commerce	OTIF, last-mile delivery, omnichannel fulfillment, seasonal demand peaks	OTIF >98%, inventory turns 8-15x, return rate <15%	Amazon, Walmart, Zara
■ Consumer Packaged Goods	Case fill rate, trade promotion management, retailer compliance, packaging ESG	Case fill rate >98%, forecast accuracy >85%, on-shelf availability >97%	P&G, Unilever, Nestle
✈️ ■ Aerospace & Defense	AS9100D compliance, long-lead parts (12-36 months), ITAR/EAR export controls	On-time delivery >95%, parts traceability 100%, zero-defect culture	Boeing, Airbus, Lockheed Martin
■ Healthcare	UDI traceability, sterile supply chain, GPO compliance, FDA 21 CFR 820	Stockout rate <0.5% critical items, recall response <24hrs, accuracy >99.5%	Medtronic, Stryker, Becton Dickinson
■ General	Standard SCOR + Gartner benchmarks across all 8 domains	Per-domain world-class thresholds as defined in Section 5	Gartner Top 25 supply chains

7. PRIVACY, LEGAL & COMPLIANCE

Illustrative & Directional Use Only

All assessments, scores, benchmarks, and financial estimates produced by this tool are for informational and directional purposes only. They do not constitute professional financial, legal, or management consulting advice. Users are encouraged to validate all outputs with qualified advisors before making business or investment decisions.

Financial data privacy

What is collected	Financial figures entered optionally in the CBA section
How it is used	Solely to generate the cost-benefit analysis during the active session. Passed to the Anthropic Claude API per their Privacy Policy.
Storage	None. No financial data is stored, logged, or retained anywhere. Data exists only in the browser session and is permanently discarded on tab close, Reset click, or session expiry.
Sharing	Never sold, shared with third parties, or used to train AI models.
Participation	Entirely voluntary. A consent checkbox must be actively checked before financial inputs appear. The full assessment runs without any financial data.
Regulatory basis	Operated in accordance with GDPR (EU) 2016/679 and the California Consumer Privacy Act (CCPA). As no personal or financial data is stored or processed beyond the active session, data subject rights (access, correction, deletion) are satisfied automatically by session expiry.

Web search & Anthropic API

This tool uses the Anthropic Claude API including the web search tool (`web_search_20250305`). All data sent to the API — including any financial context provided — is processed under Anthropic's Privacy Policy at anthropic.com/legal/privacy. Web search requires a paid Anthropic account and may incur additional per-search costs. The tool sets `max_uses=5` per assessment call to manage costs. A graceful fallback to static analysis runs automatically if web search is unavailable.

8. PROJECT STRUCTURE

```
supply-chain-health-agent/
agent.py # Agentic core – web search tool, action pack, market intel
domains.py # Domain definitions, 2025 KPI benchmarks, system prompt builder
personas.py # NEW – persona configs, executive prompt extension, customisation
verticals.py # 8 industry vertical presets
scoring.py # Score parsing and interpretation
cli.py # Interactive terminal interface
app.py # Streamlit web UI – premium SaaS design, tabbed results
requirements.txt
.env # API key (gitignored – never committed)
README.md
```

Key architectural decisions in v2.0

personas.py (new) — Single source of truth for all persona configuration — UI labels, context questions, customisation options, and the executive system prompt extension. Clean separation means Phase 2 Enterprise persona drops in without touching other files.

Web search as server tool — Uses Anthropic's server-side `web_search_20250305` — Anthropic executes searches, we just pass the tool definition. Response handling uses `_extract_text()` to join all text blocks from mixed-content responses (text + search results + `tool_use`).

Token budget by persona — Executive track gets 4,000 tokens (scenarios + roadmap need it). CBA-enabled analyst track gets 2,500. Standard analyst gets 1,500. Action pack uses 2,000.

Graceful degradation — `BadRequestError` on the main assessment call triggers an automatic retry without web search — so the app never breaks if search isn't enabled on the account.

9. QUICK-START GUIDE

Step 1 Clone the repository

```
git clone https://github.com/dwnjuguna/supply-chain-health-agent.git
cd supply-chain-health-agent
```

Step 2 Install dependencies

```
pip install -r requirements.txt
```

Step 3 Add your Anthropic API key

```
echo "ANTHROPIC_API_KEY=your-key-here" > .env
# Get your key at: console.anthropic.com
```

Step 4a Run the terminal CLI


```
python3 cli.py
```

Step 4b**Run the Streamlit web UI**

```
streamlit run app.py  
# Opens at http://localhost:8501
```

Step 5**Deploy to Streamlit Cloud (public URL)**

```
# Push to GitHub, connect at share.streamlit.io  
# Add ANTHROPIC_API_KEY under App Settings -> Secrets  
# Live at: https://your-app-name.streamlit.app
```

10. LINKS & CONTACT

■ Live app	supply-chain-health-agent.streamlit.app
■ GitHub repository	github.com/dwnjuguna/supply-chain-health-agent
■ Anthropic Claude SDK	docs.anthropic.com
■ Get an API key	console.anthropic.com
■ SCOR model reference	ascm.org/learning-development/scor
■ Anthropic privacy	anthropic.com/legal/privacy
■■ License	MIT — free to use, share, and adapt
■ Author	David Njuguna GitHub: @dwnjuguna

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Built with the Anthropic Claude SDK | github.com/dwnjuguna/supply-chain-health-agent